

ECG Anomaly Detection — Session Report

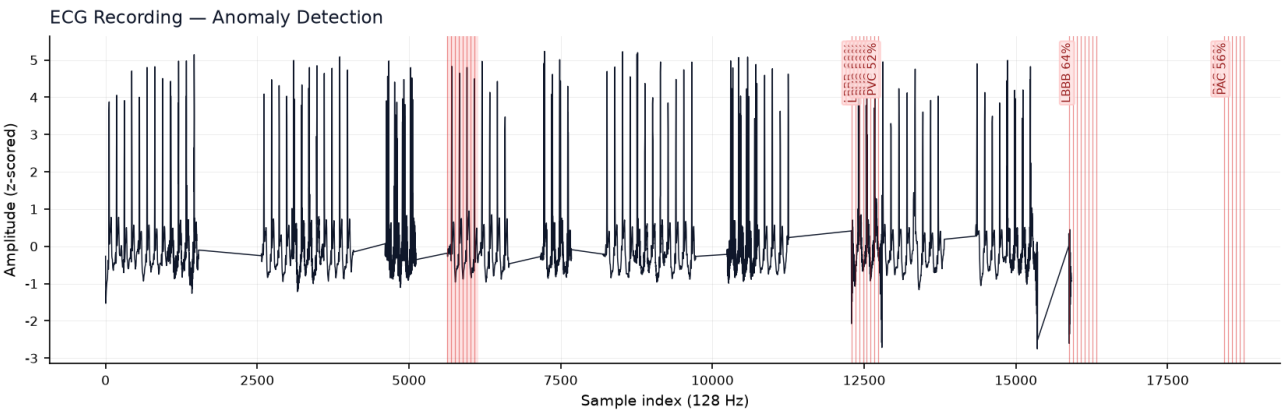
Prepared for: kk • Session ID: a3ef6613 • Generated: 2026-07-05 06:19:42

Session Information

Patient / User	Sriyantha Sadaruwan
Source	mit-bih-arrhythmia (121)
Started	2026-07-05 06:19:42
Ended	2026-07-05 06:21:01
Duration	77.6 s
Status	completed
Total beats analyzed	38,912
Anomalous beats	9,926 (25.51%)
Model	ecg-ae-v2 (v1783189189) • Dilated-U-Net-Autoencoder
Detection threshold (MAE)	0.1341

ECG Recording & Detection Results

The following plot shows the recorded ECG signal (z-score normalized). Red-shaded regions indicate samples flagged as anomalous by the autoencoder (reconstruction error exceeding the trained threshold).



Main Arrhythmia Finding

Left Bundle Branch Block was the main high-confidence arrhythmia type detected in this session. It appeared 46 time(s) with average confidence 78.4% and maximum confidence 92.0%.

Reason: this type is shown because the autoencoder first flagged the ECG segment as anomalous by reconstruction error, and the classifier then assigned this arrhythmia label with at least 50% confidence. Lower-confidence labels are excluded from this report to avoid over-interpreting uncertain model output.

Example: time 2026-07-05 06:20:21, anomaly score 0.0601, threshold 0.0568, classifier confidence 92.0%.

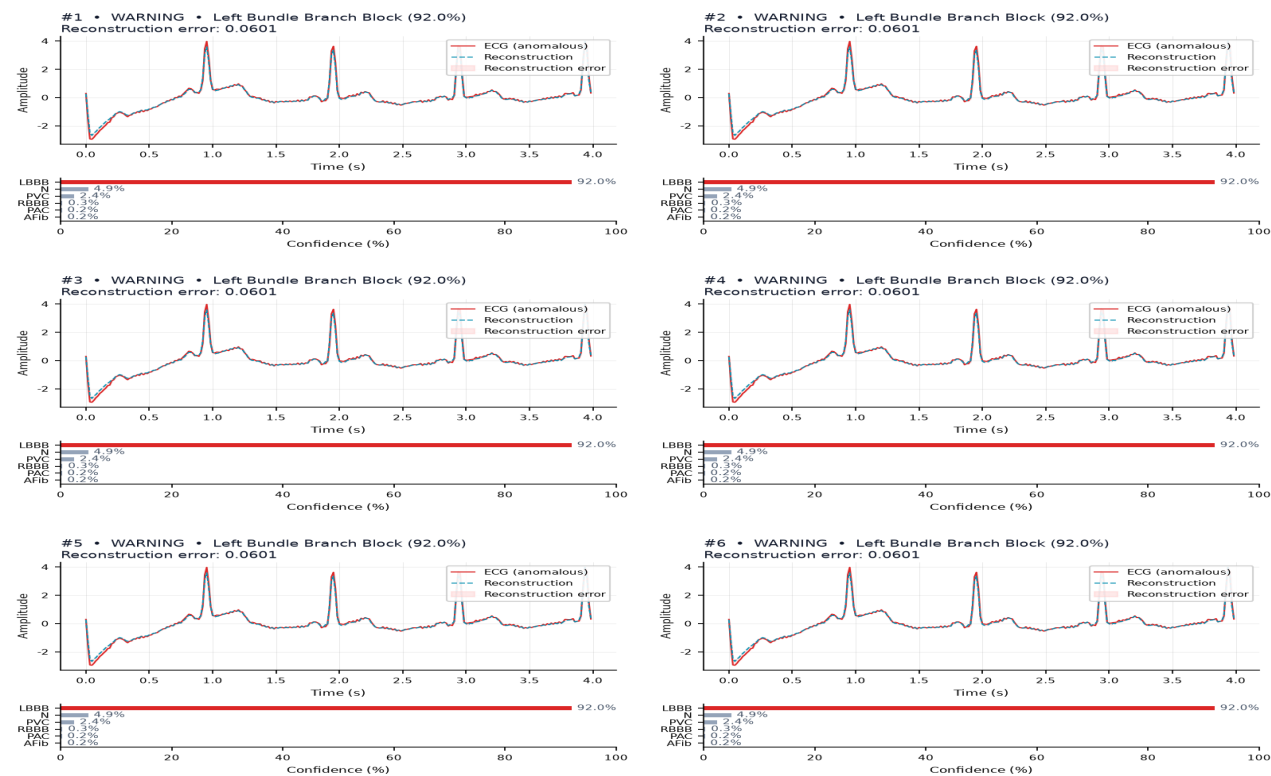
Clinical meaning: Left Bundle Branch Block (LBBB) is a condition where electrical signals to the left ventricle of the heart are delayed. This causes the left ventricle to contract later than the right ventricle, resulting in an abnormal ECG pattern with a wide QRS complex. LBBB affects the timing of the heartbeat but not necessarily the heart's ability to pump blood.

Why it matters: LBBB itself is not life-threatening, but it often indicates underlying heart disease. New-onset LBBB can sometimes signal an acute heart attack and requires immediate medical evaluation. Chronic LBBB (present for a long time) is usually monitored but may require treatment of the underlying condition.

Arrhythmia Type	Detections	Avg Confidence	Example
Left Bundle Branch Block	46	78.4%	score 0.0601, conf 92.0%
Premature Atrial Contraction	37	73.5%	score 0.0585, conf 90.0%
Premature Ventricular Contraction	7	57.0%	score 0.0610, conf 70.7%
Right Bundle Branch Block	2	55.7%	score 0.0649, conf 55.7%

Detected Anomaly Waveforms

The following plots show example ECG segments (red) that triggered alerts, alongside the autoencoder's reconstruction of what it expected to see (blue dashed). The shaded region between the two curves represents the reconstruction error that exceeded the detection threshold. When available, examples are selected from arrhythmia classifications above 50% confidence.



Clinical Summary

25.5% of analyzed samples were flagged as anomalous, suggesting sustained abnormal ECG morphology. Urgent cardiology review is recommended. High-confidence classifier findings in this report

include: Left Bundle Branch Block, Premature Atrial Contraction, Premature Ventricular Contraction, Right Bundle Branch Block. Only arrhythmia labels with confidence above 50% are included.

Physician Notes

Physician signature: _____

Date: _____

This report is generated automatically by an unsupervised deep-learning anomaly detection system and is intended for decision support only. It does not constitute a medical diagnosis and should not replace clinical judgment by a qualified physician.